A Comparison of Four Methods for Predictive Musculoskeletal Simulations of Human Walking

Supplementary Material

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Contents

1 Cost & Reward Functions

1.1 Optimal Control

For optimal control, the standard cost function has the following shape:

$$J = w_{\mathrm{COT}} J_{\mathrm{COT}} + w_{\mathrm{effort}} J_{\mathrm{effort}} + w_T J_T + w_{\ddot{q}} J_{\ddot{q}} + w_{\mathrm{lim}} J_{\mathrm{lim}} + w_{da/dt} J_{da/dt} + w_{dF/dt} J_{dF/dt},$$

where the terms represent the squared metabolic cost based on Bhargava et al. [1] (J_{COT}), squared sum of muscle activations (J_{effort}), activation of coordinate actuators (J_T), joint accelerations ($J_{\ddot{q}}$), coordinate limit torques (J_{lim}), and slack controls ($J_{da/dt}$, $J_{dF/dt}$). The weights for each term are reported in Table 1.

Table 1: Weights used inside the cost function of the predictive musculoskeletal simulations.

Symbol	**Value**
w_{COT}	500
w_{effort}	2000
w_T	1000000
$w_{\ddot{q}}$	50000
w_{lim}	1000
$w_{da/dt}$	1e-3
$w_{dF/dt}$	1e-3

1.2 Muscle-Reflex Model

For the muscle-reflex model, the standard cost function has the following shape:

$$J = w_{\text{Gait}} J_{\text{Gait}} + w_{\text{Effort}} J_{\text{Effort}} + w_a J_a + w_{\text{DOF}_{\text{ankle}}} J_{\text{DOF}_{\text{ankle}}} + \\ w_{\text{DOF}_{\text{knee}}} J_{\text{DOF}_{\text{knee}}} + w_{\text{DOF}_{\text{pelvis tilt}}} J_{\text{DOF}_{\text{pelvis tilt}}} + w_{\text{GRF}} J_{\text{GRF}},$$

where the terms represent: a gait measure to keep the model's centre of mass above 0.85 m, and its velocity above 1 m/s (J_{Gait}), the metabolic cost as defined by Wang et al. [2] (J_{effort}), cubed muscle activations (J_a), coordinate position, velocity and limit torque limits (J_{DOF}), and a penalty for ground reaction forces above 1.4 N/kg (J_{GRF}). The ankle angle is penalized when its value goes outside the $[-60°, 60°]$ range. For the knee, any limit torque is penalized, and for the pelvis tilt angles outside of $[-45°, 0°]$ and velocities outside of $[-10°, 10°]$ are penalized. A detailed description of each term and fitness parameter can be found on the official SCONE documentation website [3]. The weights for each term are reported in Table 2.

Table 2: Weights used inside the cost function of the predictive musculoskeletal simulations using a muscle-reflex model.

Symbol	**Value**
w_{Gait}	100
w_{Effort}	0.1
w_a	2000
$w_{\text{DOF}_{\text{ankle}}}$	0.1
$w_{\text{DOF}_{\text{knee}}}$	0.02
$w_{\text{DOF}_{\text{pelvis tilt}}}$	0.01
w_{GRF}	10

1.3 Central Pattern Generator

For the central pattern generator model, the cost function has largely the same shape as the muscle-reflex model, with some updated weights (Table 3):

$$J = w_{\text{Gait}} J_{\text{Gait}} + w_{\text{Alive}} J_{\text{Alive}} + w_{\text{Effort}} J_{\text{Effort}} + w_a J_a + w_{\text{DOF}_{\text{ankle}}} J_{\text{DOF}_{\text{ankle}}} + w_{\text{DOF}_{\text{knee}}} J_{\text{DOF}_{\text{knee}}} + w_{\text{DOF}_{\text{pelvis tilt}}} J_{\text{DOF}_{\text{pelvis tilt}}} + w_{\text{GRF}} J_{\text{GRF}}$$

The additional term (J_{Alive}), inspired by [4], adds a penalty depending on the distance travelled by the model (t_x), and the time at which the model falls (t), compared to the final time of the simulation (t_{end}) and the initial centre of mass position (t_{x_0}):

$$J_{\text{Alive}} = |t - t_{end}| + \max\left(\frac{1}{t_{end}}, \frac{1}{|t_x - t_{x_0}|}\right).$$

The weights for each term are reported in Table 3.

Initially, we also added a mimic measure, tracking the kinematics and muscle activations of the muscle-reflex solution from SCONE. The weights used for these optimizations are specified inside of Table 4.

Table 3: Weights used inside the cost function of the predictive musculoskeletal simulations using a central pattern generator model.

Symbol	**Value**
w_{Gait}	1
w_{Alive}	10
w_{Effort}	1
w_a	2000
$w_{\text{DOF}_{\text{ankle}}}$	0.1
$w_{\text{DOF}_{\text{knee}}}$	0.02
$w_{\text{DOF}_{\text{pelvis tilt}}}$	0.1
w_{GRF}	10

Table 4: Weights used inside the cost function of the predictive musculoskeletal simulations using a central pattern generator model, with a tracking term.

Symbol	**Value**
w_{Gait}	100
w_{Alive}	10
w_{Effort}	1
w_a	2000
$w_{\text{DOF}_{\text{ankle}}}$	0.1
$w_{\text{DOF}_{\text{knee}}}$	0.02
$w_{\text{DOF}_{\text{pelvis tilt}}}$	0.01
w_{GRF}	10
w_{track}	500

1.4 Deep Reinforcement Learning

The reward function for deep reinforcement learning has been taken from [5], without any changes. The reward consists of the following terms:

$$r = w_{\text{vel}} r_{\text{vel}} - c_{\text{effort}} - c_{\text{pain}},$$

where the first term r_{vel} encourages the model to walk at a desired velocity ($1.2\ \text{m/s}$):

$$r_{vel} = \begin{cases} \exp\left[-(v - v_{\text{target}})^2 \text{ if } v < v_{\text{target}}\right] \\ 1 \text{ otherwise,} \end{cases}$$

c_{effort} penalizes cubed muscle activations and encourages smooth muscle activations:

$$c_{\text{effort}} = \alpha(t) \sum_{m=1}^{M} a_m^3 + w_{\text{smooth}} \frac{1}{M} \sum_{m=1}^{M} (u_m - u_m^{\text{prev}})^2 + w_{\text{nmuscle}} \sum_{m=1}^{M} \left[a_m > 0.15\right],$$

and c_{pain} penalizes large ground reaction forces and joint limit torques:

$$c_{\text{pain}} = w_{\text{joint limit}} \sum_{j=1}^{J} \tau_i^{\text{lim}} + w_{\text{GRF}} \max\left(0, \frac{1}{W} \sum_{k=1}^{2} F_{\text{GRF}} - 1.2\right).$$

A summary of the weights can be found in Table 5.

Table 5: Reward keys used inside the cost function of the predictive musculoskeletal simulations using deep reinforcement learning. A detailed description of the reward function can be found in [5].

Symbol	**Value**
w_{vel}	10
w_{GRF}	-0.07281
$w_{\text{joint limit}}$	-0.1307
w_{smooth}	-0.097
w_{nmuscle}	-1.57929

2 Metrics

2.1 Sagittal Plane Kinematics

Table 6: Average root-mean-square errors, over the left and right side, computed for the different methods, for the sagittal plane kinematics, compared to data from [6].

Method	**hip (avg)** [°]	**knee (avg)** [°]	**ankle (avg)** [°]	**total avg** [°]
MR	5.93	4.15	3.87	4.65
CPG	9.66	7.54	3.95	7.05
OC	4.25	5.14	8.75	6.04
DRL	9.66	18.25	10.69	12.86
DRL (H0918v3)	4.63	15.35	5.97	8.65

Table 7: Average correlation coefficients, over the left and right side, computed for the different methods, for the sagittal plane kinematics, compared to data from [6].

Method	**hip (avg)** [-]	**knee (avg)** [-]	**ankle (avg)** [-]	**total avg** [-]
MR	0.98	0.98	0.86	0.94
CPG	0.90	0.96	0.83	0.90
OC	1.00	0.96	0.52	0.83
DRL	0.80	0.56	0.60	0.65
DRL (H0918v3)	0.97	0.77	0.76	0.83

Table 8: Average experimental match, over the left and right side, computed for the different methods, for the sagittal plane kinematics, compared to data from [6].

Method	**hip (avg)** [%]	**knee (avg)** [%]	**ankle (avg)** [%]	**total avg** [%]
MR	88.62	86.14	76.74	83.83
CPG	67.33	56.44	79.71	67.83
OC	99.51	74.76	40.10	71.45
DRL	56.44	10.89	28.22	31.85
DRL (H0918v3)	90.60	59.41	52.48	67.49

2.2 Ground Reaction Forces

Table 9: Average root-mean-square errors, over the left and right side, computed for the different methods, for the sagittal plane kinematics, compared to data from [6].

Method	GRFx (avg) [N/kg]	GRFy (avg) [N/kg]	total avg [N/kg]
MR	0.23	0.81	0.52
CPG	0.34	1.31	0.82
OC	0.44	1.77	1.11
DRL	0.40	1.45	0.93
DRL (H0918v3)	0.60	1.88	1.24

Table 10: Average correlation coefficients, over the left and right side, computed for the different methods, for the sagittal plane kinematics, compared to data from [6].

Method	GRFx (avg) [-]	GRFy (avg) [-]	total avg [-]
MR	0.96	0.98	0.97
CPG	0.93	0.96	0.94
OC	0.86	0.93	0.89
DRL	0.89	0.94	0.91
DRL (H0918v3)	0.75	0.90	0.83

Table 11: Average experimental match, over the left and right side, computed for the different methods, for the sagittal plane kinematics, compared to data from [6].

Method	GRFx (avg) [%]	GRFy (avg) [%]	total avg [%]
MR	76.24	74.26	75.25
CPG	73.77	65.35	69.56
OC	49.01	69.31	59.16
DRL	61.39	48.52	54.95
DRL (H0918v3)	52.48	56.44	54.46

2.3 Muscle Activations

Table 12: Average correlation coefficients, over the left and right side, computed for the different methods, for the sagittal plane kinematics, compared to data from [6].

Method	**ham** [-]	**gastroc** [-]	**tibant** [-]	**vasti** [-]	**rectfem** [-]	**total avg** [-]
MR	-0.05	0.63	0.75	0.41	-0.32	0.28
CPG	-0.34	0.84	0.22	0.66	-0.22	0.23
OC	0.45	0.85	0.57	0.68	-0.18	0.47
DRL	0.46	0.50	0.18	0.43	-0.04	0.31
DRL (H0918v3)	0.66	0.64	0.20	0.90	0.14	0.50

3 Computation Time

The simulation time is difficult to compare as the simulations ran on different hardware, and since the MR, CPG and DRL methods were accelerated using the HyFyDy software. OC gave the fastest results ($\pm 1 \text{ min}$), while DRL took the longest to train ($11 \cdot 10^7$ steps, ± 1 day). For the MR and, especially, the CPG method, multiple simulations were required to get a stable solution. However, HyFyDy noticeably reduced the simulation time, allowing more exploration of different settings, like seeds, or cost function weights.

4 References

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